

# INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



## Assignment Report

### M.Tech in Integrated Circuit Design and Technology (ICDT)

|                         |                             |
|-------------------------|-----------------------------|
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| <b>Subject Name:</b>    | CMOS Analog IC Design       |
| <b>Subject Code:</b>    | EE 651                      |
| <b>Assignment No:</b>   | 03                          |
| <b>Submission Date:</b> | 05 September 2025           |

## Question 1

Common Source Amplifier Design (Question 1) Design a Common Source (CS) amplifier with active load to meet the following specifications: Design a Common Source (CS) amplifier with active load to meet the following specifications:

- Voltage Gain greater  $> 100\text{V/V}$
  - Power dissipation  $< 10\mu\text{W}$
  - Minimum input voltage swing  $= 100\mu\text{V}$
  - Maximum input voltage swing  $= 12\text{mV}$
- (a) Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
- (b) Plot the frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.

## Schematic

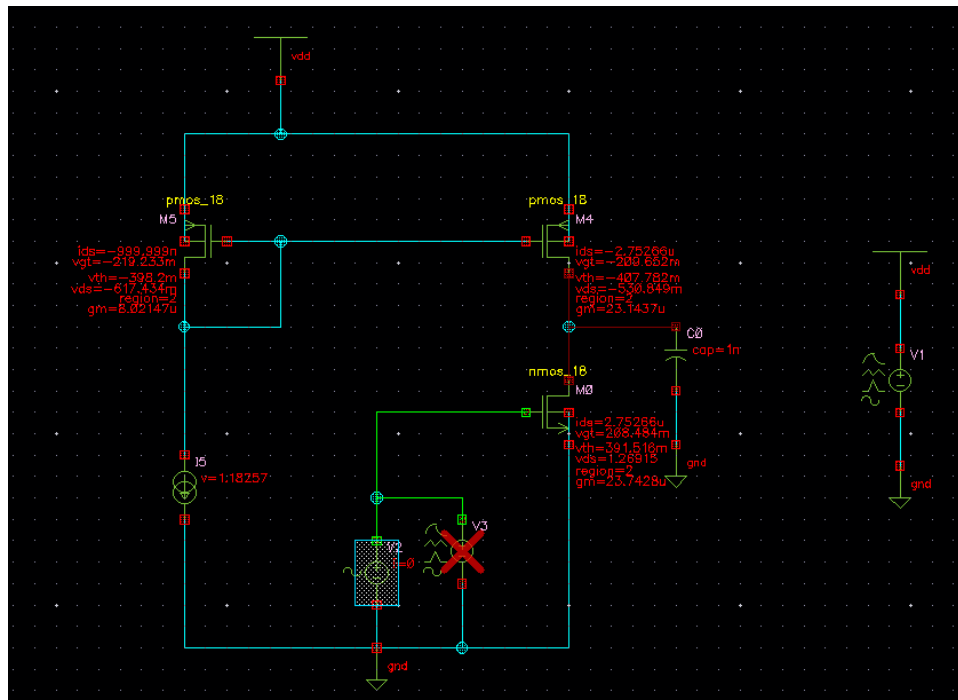


Figure 1: Common Source Schematic Amplifier with Active Load

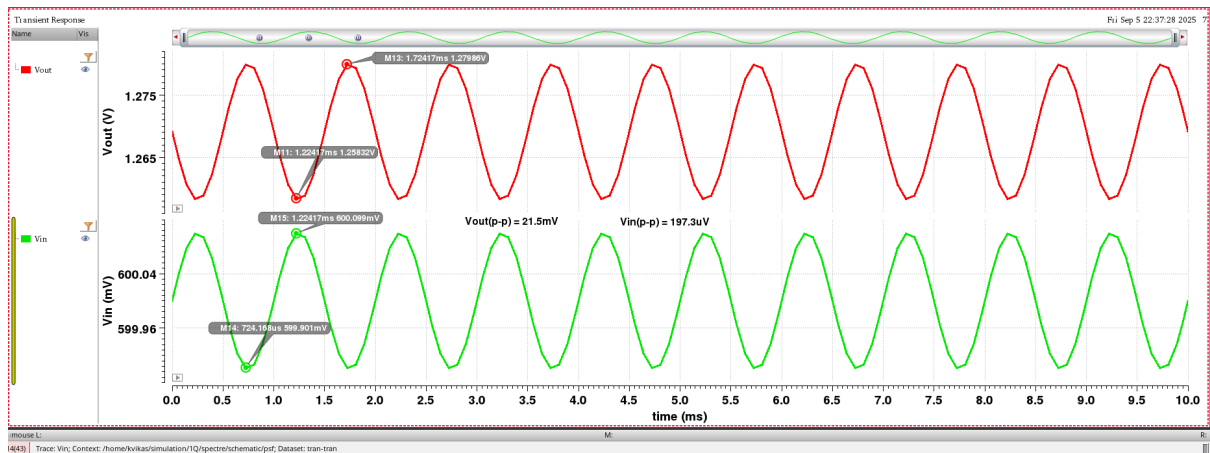


Figure 2: Output Swing

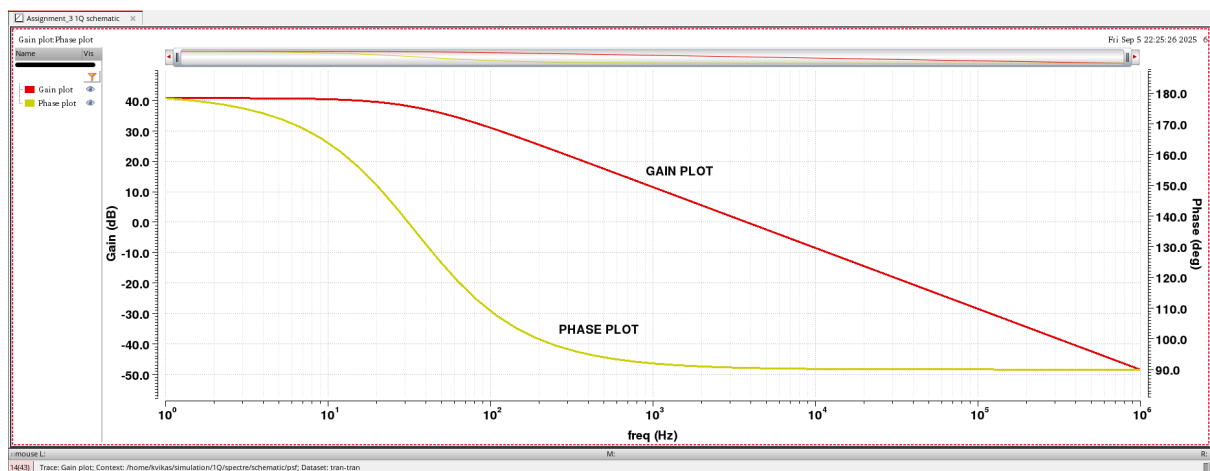


Figure 3: Gain and Phase Plot

## results

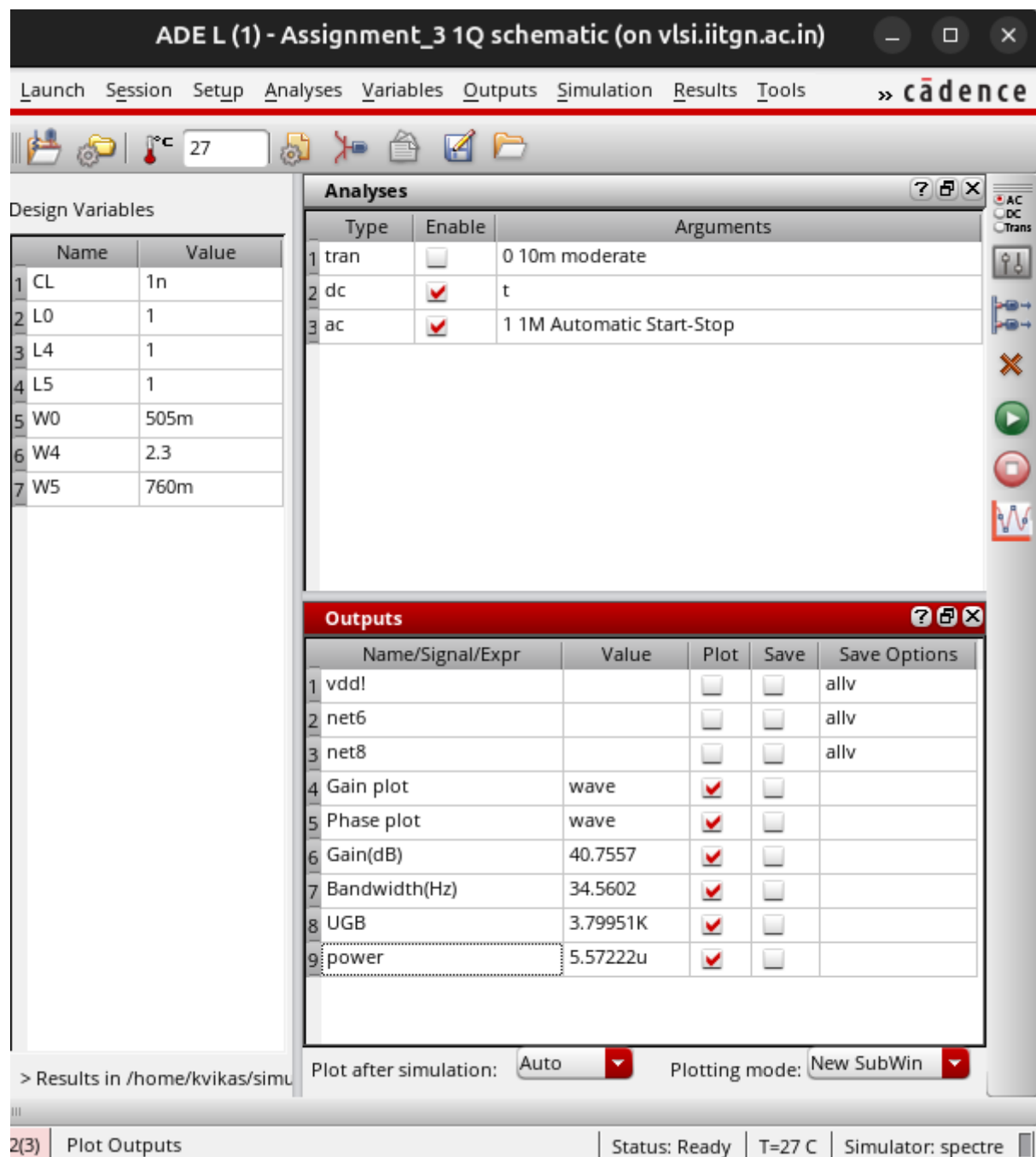


Figure 4: Outputs

## Observations

- For the schematic in (Fig. 6) the  $V_{out} = 21.5mV$  for  $V_{in} = 197.3mV$ .
- $\frac{V_{out}}{V_{in}} = 108.9$
- Therefore the signal is amplified.
- Gain = 40.7dB
- Bandwidth = 34.56Hz
- Unit Gain Bandwidth = 3.79KHz
- power = 5.57uW

Design a Common Drain (CD) amplifier with active load for the following specification:

- Voltage gain:  $> 0.95 \text{ V/V}$
  - Power dissipation:  $< 2 \text{ mW}$
  - Output impedance ( $Z_{\text{out}}$ ):  $< 100 \Omega$
  - Minimum input voltage swing:  $20 \text{ mV}$
  - Maximum input voltage swing:  $1 \text{ V}$
- (a) Calculate the output swing voltage and gain for the minimum and maximum input swings using transient analysis, and also plot the transient response.
- (b) Plot frequency response (both Gain and Phase response) of the amplifier. Extract Bandwidth and Unity Gain Frequency (UGF) and explain the results.

Figure 5: Common Drain Schematic Amplifier with Active Load

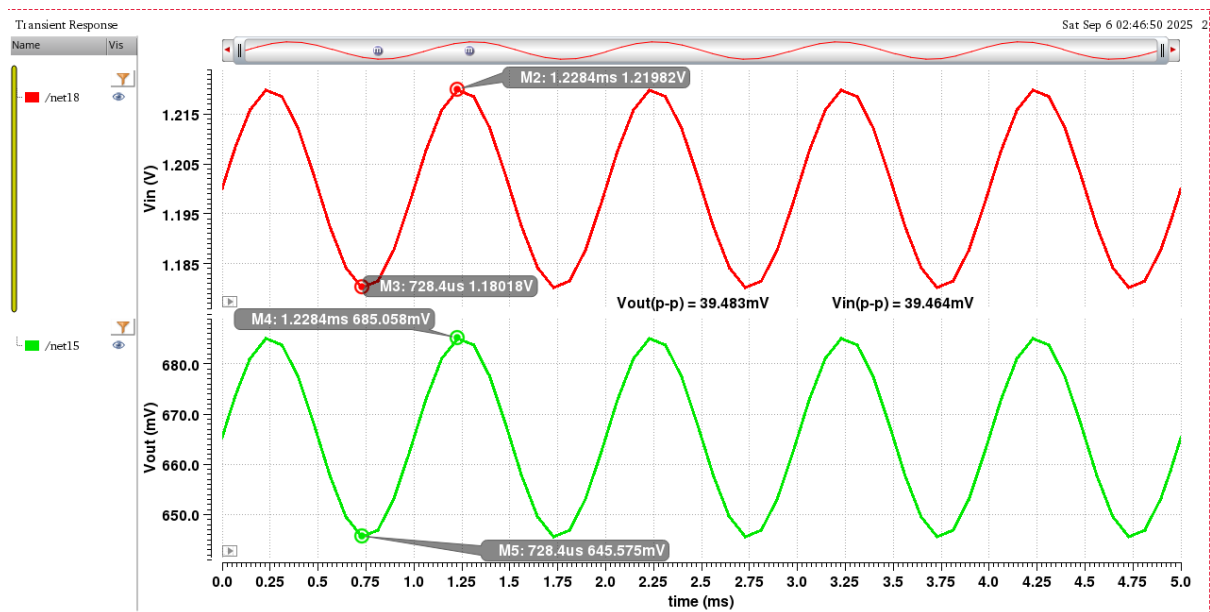


Figure 6: Output Swing

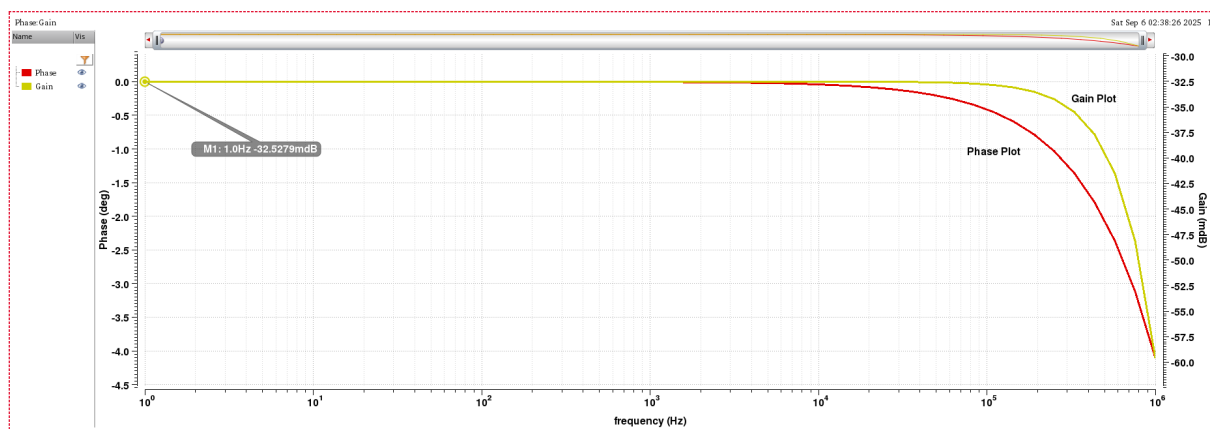


Figure 7: Gain and Phase Plot

## results

Analyses

?

|   | Type | Enable                              | Arguments                 |
|---|------|-------------------------------------|---------------------------|
| 1 | dc   | <input checked="" type="checkbox"/> | t                         |
| 2 | ac   | <input checked="" type="checkbox"/> | 1 1M Automatic Start-Stop |
| 3 | tran | <input type="checkbox"/>            | 0 5m moderate             |
|   |      |                                     |                           |

Outputs

?

|   | Name/Signal/Expr | Value     | Plot                                | Save                     | Save Options |
|---|------------------|-----------|-------------------------------------|--------------------------|--------------|
| 1 | net18            |           | <input checked="" type="checkbox"/> | <input type="checkbox"/> | allv         |
| 2 | net15            |           | <input checked="" type="checkbox"/> | <input type="checkbox"/> | allv         |
| 3 | Phase            | wave      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |
| 4 | Gain             | wave      | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |
| 5 | gain(dB)         | -32.5302m | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |
| 6 | power            | 123.982u  | <input checked="" type="checkbox"/> | <input type="checkbox"/> |              |
|   |                  |           |                                     |                          |              |

Figure 8: Outputs

## Observations

- For the schematic in (Fig. 6) the  $V_{out} = 39.483mV$  for  $V_{in} = 39.64mV$ .
- Therefore  $\frac{V_{out}}{V_{in}} = 0.996$
- Therefore, we receive almost the same input signal at the output.
- Gain = -32.53mdB
- power = 123.982uW